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Homework # 1. 420.

1). A real-valued function $f(x)$ defined on the real line is called an even function if $f(t) = f(-t)$, $\forall t \in R$. Prove that the set of even functions on the real line with the usual operations of addition and scalar multiplication is a vector space.

Proof.

Let $\mathcal{F}(R)$ denote the set of all even functions with the operations of addition and scalar multiplication:

$$\mathcal{F} = \mathcal{F}(R) = \{ f : R \to R \mid f(t) = f(-t), \forall t \in R \} \quad (1)$$

VS 1:

$\forall f, g \in \mathcal{F}$, $\overbrace{f+g}^{\in \mathcal{F}} = \overbrace{g+f}^{\in \mathcal{F}}$ (commutativity)

holds. No problem.

VS2:

$\forall f, g, h \in \mathcal{F}$ $\underbrace{(f+g)+h}_{\in \mathcal{F}} = \underbrace{f+(g+h)}_{\in \mathcal{F}}$ (associativity)

holds as well

VS 3:

$\exists\, 0 \in \mathcal{F}$ s.t. $f + 0 = f$, $\forall f \in \mathcal{F}$,

since $0(t) = 0(-t) = 0$ - zero function always.

VS 4:

$\forall f \in \mathcal{F}$, $\exists\, g \in \mathcal{F}$ s.t.

$$f + g = 0 \implies g = \underbrace{\overset{\in R}{(-1)} \cdot \overset{\in \mathcal{F}}{f}}_{\in \mathcal{F}}$$

scalar multiplication is defined on the space. That's been given.

VS 5:

$\forall f \in \mathcal{F}$, $1 \cdot f = f \in \mathcal{F}$ holds.

VS 6:

$\forall a, b \in R$ and $\forall f \in \mathcal{F}$

$(ab) f = a \cdot (bf)$ holds since scalar multiplication has been defined on the space.

VS 7:

$\forall a \in R$, $\forall f, g \in \mathcal{F}$

$a \cdot (f + g) = (a \cdot f + a \cdot g) \in \mathcal{F}$ holds

and finally

VS 8:

$\forall a, b \in R$, $\forall f \in \mathcal{F}$

$(a+b) f = (af + bf) \in \mathcal{F}$ holds.

Hence, $\mathcal{F} = \mathcal{F}(R)$ space is a vector space.

2) Let V denote the set of ordered pairs of real numbers. If $x = (x_1, x_2) \in V$, $y = (y_1, y_2) \in V$ and $c \in F$, define

$$x + y = (x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_1 \cdot y_1) \quad (1)$$

$$cx = c(x_1, x_2) = (cx_1, cx_2) \quad (2)$$

Is V a vector space over R?

Solution: [Preamble]

VS1:

$\forall x = (x_1, x_2) \in V$, $\forall y = (y_1, y_2) \in V$, $x + y =$

$$= (x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_1 \cdot y_1) \quad (3)$$

$$y+x = (y_1, y_2) + (x_1, x_2) = (y_1 + x_1, y_1 \cdot x_1) \quad (4)$$

$(3) \equiv (4) \Rightarrow$ commutativity holds.

VS 2:

$\forall x \in (x_1, x_2), \forall y = (y_1, y_2), \forall z = (z_1, z_2) \in V$

$$(x+y)+z = (x_1 + y_1 + z_1, x_1 \cdot y_1 \cdot z_1) \quad (5)$$

$$x+(y+z) = (x_1 + y_1 + z_1, x_1 \cdot y_1 \cdot z_1) \quad (6)$$

$\Rightarrow$

Associativity holds as well, as $(5) \equiv (6)$.

VS 3:

$\exists\, 0 \in V$ s.t. $\forall x = (x_1, x_2) \in V$

$$x + 0 = \underbrace{(x_1, x_2) + 0 = (x_1, x_2)}, \quad (7) \Rightarrow$$

$$0 = (x_1, x_2) - (x_1, x_2) = (x_1, x_2) + \underbrace{(-x_1, -x_2)}_{\text{(x works has been given}}$$

$$0 = (x_1, x_2) + (-x_1, -x_2)$$

$$(x_1, x_2) + (-x_1, -x_2) = (\underbrace{x_1 + (-x_1)}_{0}, \underbrace{x_1 \cdot (-x_1)}_{0})$$

$$\begin{cases} x_1 + (-x_1) = 0 \\ x_1 \cdot (-x_1) = 0 \end{cases} \Rightarrow \begin{cases} x_1 \in R \\ x_1 = 0 \end{cases}, \quad \text{Yet}$$

(7) should work $\forall x = (x_1, x_2) \in V$, not just for $x = (0,0)$ point. V <u>Fails</u> to be a vector space.

VS4:

$\forall x = (x_1; x_2) \in V \qquad \exists y = (y_1, y_2) \in V$

$x + y = 0 \quad \Rightarrow$

$(x+y) = (x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_1 \cdot y_1) = 0 \quad (8)$

$$\begin{cases} x_1 + y_1 = 0 \\ x_1 \cdot y_1 = 0 \end{cases} \Rightarrow \quad y_1 = -x_1$$

$x_1 \cdot y_1 = 0$

$x_1 \cdot (-x_1) = 0$

$-x_1^2 = 0 \quad \Rightarrow$

$x_1 = 0 \qquad (8)$

Yet, (8) should work $\forall x = (x_1, x_2) \in V$ not just for $x = (0,0)$! V <u>Fails</u>.

Vs 5:

$\forall x = (x_1, x_2) \in V \qquad 1 \cdot x = x$ holds as

$1 \cdot (x_1, x_2) = (1 \cdot x_1, 1 \cdot x_2) = x.$

VS6:

$\forall a, b \in R$, $\forall x = (x_1, x_2) \in V$

$(a\,b)\,x = a(bx)$ holds.

No problem with scalar multiplication.

VS7:

$\forall a \in R$, $\forall x = (x_1, x_2)$, $\forall y = (y_1, y_2) \in V$

$a(x+y) = a((x_1, \quad) + (y_1, y_2)) = a(x_1 + y_1, x_1 y_1) =$

scalar multip. works

$= (a x_1 + a y_1, a x_1 y_1) = (a x_1, a x_2) + (a y_1, a y_2) =$

$= ax + ay$ holds as well.

VS 8:

$\forall a, b \in R$, $\forall x = (x_1, x_2) \in V$

$(a+b)\,x = ((a+b)\,x_1, (a+b)\,x_2) =$

$= (a x_1 + b x_1, a x_2 + b x_2) \neq (a x_1 + b x_1, a x_1 \cdot b x_1) =$

$= (a x_1, a x_2) + (b x_1, b x_2) = ax + bx \Rightarrow$

V fails in VS 8.

Hence, V is not a vector space over R.

3). Prove that the set of $n \times n$ upper triangular matrices is a subspace of the $M_{n \times n}(R)$

Solution:

Let $\mathcal{U}$ be the set of all $n \times n$ upper triangular matrices.

Theorem 1.3.

Let V be a vector space and W a subset of V. Then W is a subspace of V if and only if the following three conditions hold for the operations defined in V:

(i) $0 \in W$

(ii) $(x+y) \in W$ whenever $x, y \in W$

(iii) $cx \in W$ whenever $c \in F$ and $x \in W$.

Let's check $\mathcal{U}$ throug these (i) - (iii) tests:

① The zero matrix is upper triangular matrix.

$$[0]_{n \times n} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & & & \ddots & \vdots \\ 0 & & & & 0 \end{bmatrix} \in \mathcal{U} \qquad (1)$$

(i) holds for $\mathcal{U}$. ✓

② $\forall X, Y \in \mathcal{U}$

$$X + Y = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ & x_{22} & \cdots & x_{2n} \\ & & \ddots & \\ 0 & & & x_{nn} \end{bmatrix} + \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ & y_{22} & \cdots & y_{2n} \\ & & \ddots & \\ 0 & & & y_{nn} \end{bmatrix} =$$

$$= \begin{bmatrix} (x_{11}+y_{11}) & (x_{12}+y_{12}) & \cdots & (x_{1n}+y_{1n}) \\ & (x_{22}+y_{22}) & \cdots & (x_{2n}+y_{2n}) \\ & & \ddots & \vdots \\ 0 & & & (x_{nn}+y_{nn}) \end{bmatrix} \in \mathcal{U} \qquad (2)$$

(ii) holds for $\mathcal{U}$. ✓

and finally,

③ $\forall c \in R \quad \forall X \in \mathcal{U}$

$$cX = \begin{bmatrix} cx_{11} & cx_{12} & \cdots & cx_{1n} \\ & cx_{22} & \cdots & cx_{2n} \\ & & \ddots & \\ 0 & & & cx_{nn} \end{bmatrix} \in \mathcal{U} \qquad (3)$$

(iii) holds for $\mathcal{U}$ as well.

Hence, $\mathcal{U}$ is a subspace of $M_{n\times n}(R)$

4) Let $\mathcal{U}$ and W be subspaces of a vector space V. Prove that $\mathcal{U} \cup W$ is a subspace of V if and only if $\mathcal{U} \subseteq W$ or $W \subseteq \mathcal{U}$.

Solution:

① ($\Rightarrow$)

Let $\mathcal{U}$ & W be subspaces of V and assume $(\mathcal{U} \cup W)$ is a subspace of V.

We need to show (WTS) $\mathcal{U} \subseteq W$ or $W \subseteq \mathcal{U}$.

Let me denote these 2 statements as following:

$A =$ " $\mathcal{U} \cup W$ is a subspace of V " (1)

$B =$ " $\mathcal{U} \subseteq W$ or $W \subseteq \mathcal{U}$ " (2) for $\mathcal{U}, W$ which are subspaces of V.

So, WTS that $A \Rightarrow B$. (3)

But instead let me try $\bar{A} \Leftarrow \bar{B}$ (4)

for contrapositive proof, by $\bar{A}, \bar{B}$ I mean

$\bar{A} = \{ \mathcal{U} \cup W$ is NOT a subspace of $V \}$ (5)

$\bar{B} = \{$ Neither $\mathcal{U} \subseteq W$ nor $W \subseteq \mathcal{U} \}$ (6)

So, I am going to show that $\bar{B} \Rightarrow \bar{A}$.
Suppose $\bar{B}$ holds, so

Neither $\mathcal{U} \subseteq W$ nor $W \subseteq \mathcal{U}$ (7)

Take $\forall u \in \mathcal{U} \nsubseteq W \Rightarrow u \notin W$ (8) and
$\forall w \in W \nsubseteq \mathcal{U} \Rightarrow w \notin \mathcal{U}$ (9)

Since, $u \in \mathcal{U} \subseteq (\mathcal{U} \cup W)$ and $w \in W \subseteq (\mathcal{U} \cup W)$

$(u + w) \in (\mathcal{U} \cup W)$ (10) for $\forall u \in (\mathcal{U} \cup W)$
$\forall w \in (\mathcal{U} \cup W)$

which means $(\mathcal{U} \cup W)$ is a subspace.

To be honest, I messed up here, I was supposed to show $(\mathcal{U} \cup W)$ is <u>NOT</u> a subspace, but I am always coming to subspace.
If I could show this part $(\Rightarrow)$, the I would be going the other way around $(\Leftarrow)$.